

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2018

Mathematics

Paper 1

Higher Level

Time: 2 hours, 30 minutes

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 13 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

(Suggested maximum time: 5 minutes)

- | | |
|---------|--|
| Answer: | |
| Reason: | |

- [illegible]

- [illegible]

- | | |
|---------|--|
| Answer: | |
| Reason: | |

(Suggested maximum time: 5 minutes)

- [illegible]

- [illegible]

(Suggested maximum time: 15 minutes)

-
- A Venn diagram illustrating the relationships between three sets: A, B, and C. The universal set is denoted by U .
- Set A contains elements $e.$ and $d.$.
 - Set B contains elements $g.$ and $h.$.
 - Set C contains elements $i.$ and $j.$.
 - The intersection of A and B contains element $f.$.
 - The intersection of A and C contains element $i.$.
 - The intersection of B and C contains element $j.$.
 - The intersection of all three sets A, B, and C contains element $k.$.
 - Elements $l.$, $m.$, and $n.$ are located outside all three sets.

(i) List the elements of $(A \cup C) \setminus B$.

[illegible]

- [illegible]

- [illegible]

- | Statement | Tick one box only, for each statement | |
|---|--|-------|
| | True | False |
| $B \cup (A \cup C) = (B \cup A) \cup (B \cup C)$ | | |
| $A \cap (B \cap C) = (A \cap B) \cap (A \cap C)$ | | |
| $A \setminus (B \setminus C) = (A \setminus B) \setminus (B \setminus C)$ | | |

- (b) 100 people were surveyed on the different media outlets they use to keep up to date with news. They were asked which of the following they had used during the previous week: websites (W), television (T) or newspapers (N).

78 had used websites

63 had used television

44 had used newspapers

7 hadn't used any of the media outlets listed.

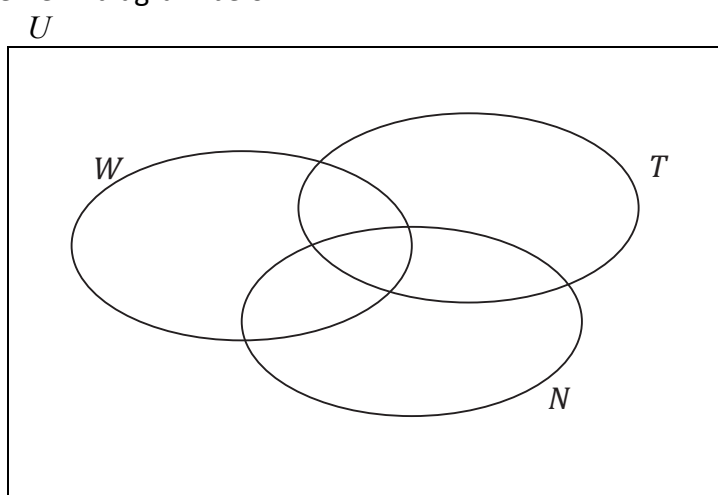
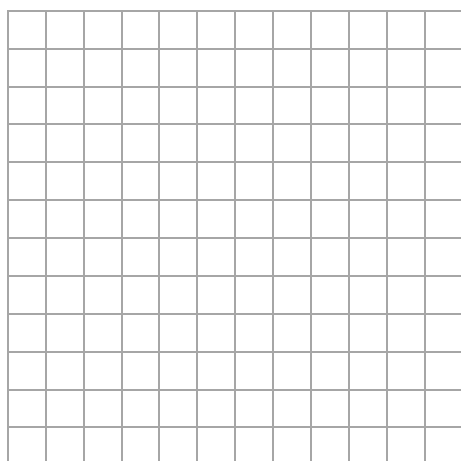
One quarter of those surveyed had used all three media outlets.

3 had used television and newspapers, but not websites.

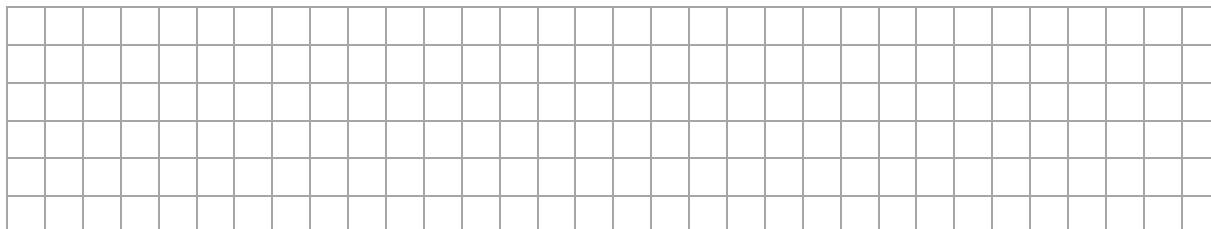
55 had used websites and television.

67 had used at least two of the media outlets listed.

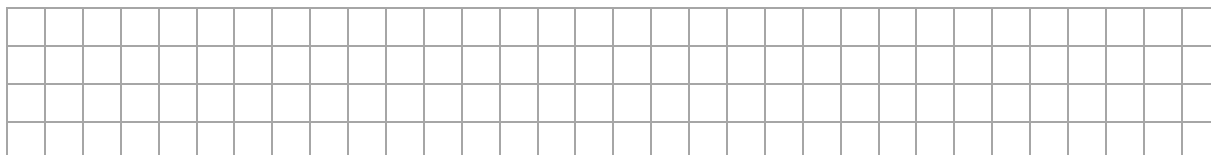
- (i) Use this information to fill in the Venn diagram below.



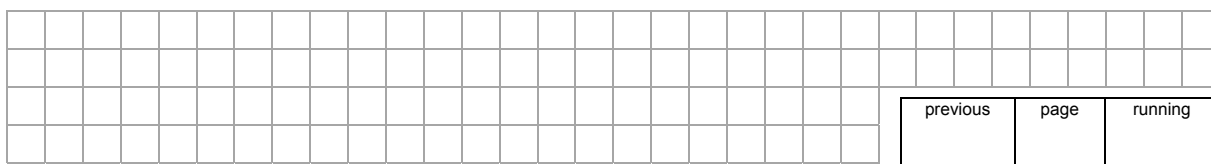
- (ii) Find the **probability** that a person chosen at random from the group had used television only during the previous week.



- (iii) Find the **probability** that a person chosen at random from the group had used at least one of the media outlets during the previous week.



- (iv) Find the **probability** that a person chosen at random from the group had not used television during the previous week.



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Question 4

(Suggested maximum time: 15 minutes)

- (a)** Write each of the following numbers in the form 2^k , where $k \in \mathbb{Q}$.

(i) 4

[illegible]

(ii) 16

[illegible]

(iii) $\sqrt{2}$

[illegible]

- (b)** Using the laws of indices, solve for x

$$4^x = \frac{16}{\sqrt{2}}.$$

[illegible]

- (c) (i) The speed of sound is 343 m/s. How many metres will sound travel in one hour? Give your answer in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$.

[illegible]

- (ii) The distances in space are much greater than what we are used to on Earth. Given that 1 astronomical unit = 149 597 871 km, how long would it take for sound to travel 1 astronomical unit?
Give your answer correct to the nearest hour.

[illegible]

Question 5 (Suggested maximum time: 10 minutes)

Question 5 (Suggested maximum time: 10 minutes)

- (a)** Factorise fully each of the following expressions.

(i) $15cd - 3de$

[illegible]

(ii) $om + np - no - pm$

[illegible]

- (b) (i)** Factorise $2x^2 - 5x - 12$.

[illegible]

(ii) Hence, or otherwise, solve the equation $2x^2 - 5x - 12 = 0$.

[illegible]

Question 6 (Suggested maximum time: 5 minutes)

Question 6 (Suggested maximum time: 5 minutes)

Solve the following equation

$$\frac{2x+5}{2} - \frac{3x-5}{5} = \frac{5}{2}.$$

[illegible]

Question 7

(Suggested maximum time: 15 minutes)

A house was bought for €145 000. The new owner spent €40 000 renovating the house before selling it for €220 000. The estate agent charged a fee of 1.5% of the selling price for selling the house.



- (a)** Find the **fee** charged by the estate agent.

[illegible]

- (b)** Find the **percentage profit** made by the owner, to one decimal place.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger margin on the left side for writing.

The profit made on selling the house was invested in a bank which offers an annual compound interest rate of 3% for two years.

- (c)** Calculate the **value** of the investment at the end of the two years.

[illegible]

- (d) No money was withdrawn, and the investment was worth €34 891.67 at the end of the third year. Calculate the **annual rate of interest** for the third year. Give your answer as a percentage, correct to one decimal place.

A large grid of 20 columns and 10 rows, intended for drawing. The grid is composed of thin black lines forming a series of small squares.

Question 8 (Suggested maximum time: 5 minutes)

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Transition year students are performing a school play.
Tickets cost €7 for adults and €3 for students and children.

On the opening night, they sold 299 tickets for a total of €1481.



- (a)** Write down two equations to represent the above information.

[illegible]

- (b)** How many adult tickets were sold?

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

- (c)** How many student and child tickets were sold?

[illegible]

(Suggested maximum time: 15 minutes)

-
- A right triangle is shown with a vertical leg of length $5 - \sqrt{3}$, a horizontal leg of length $15 + \sqrt{3}$, and a hypotenuse of length x . A right angle symbol is located at the vertex where the two legs meet.

- [illegible]

-
- A large grid of 20 columns and 10 rows, intended for drawing.

- $$5 - \sqrt{3}$$

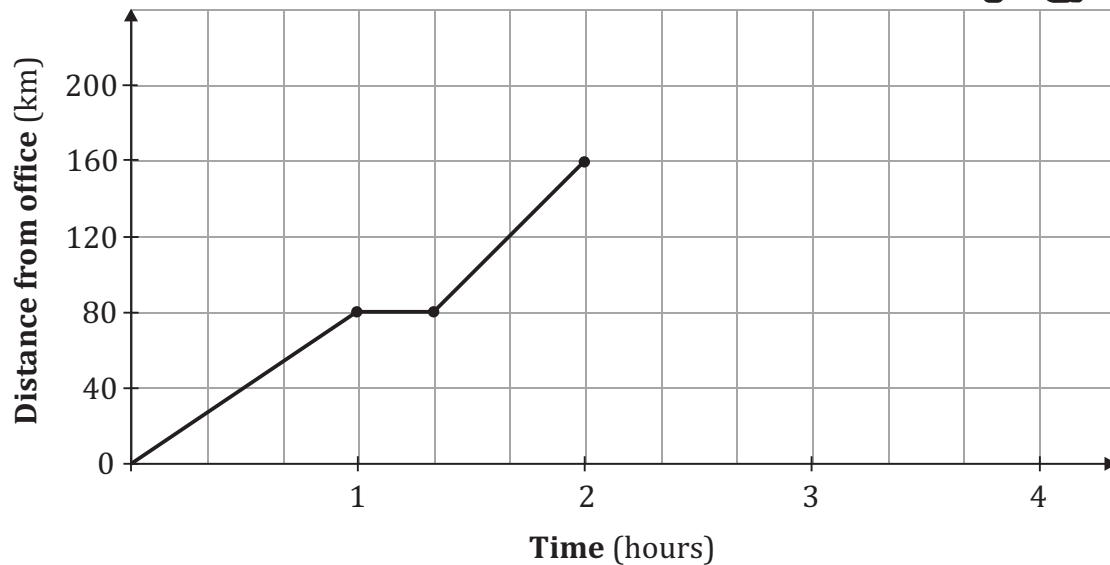
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A large grid of 20 columns and 10 rows, intended for drawing.

Question 10

(Suggested maximum time: 10 minutes)

The travel graph below shows Paul's journey by car, from his office to a meeting, on a particular day.



- (a)** Paul stopped for a break during the journey. How long was the break?

[illegible]

- (b)** Find Paul's average speed, in km/hr, after he took the break.

[illegible]

Emma left the same office and drove to another meeting at an average speed of 60 km/hr for two hours. The meeting lasted 40 minutes. She then returned to the office at twice the speed.

- (c)** Represent Emma's journey on the graph above.

- (d)** Find Emma's average speed, in km/hr, for the entire journey.

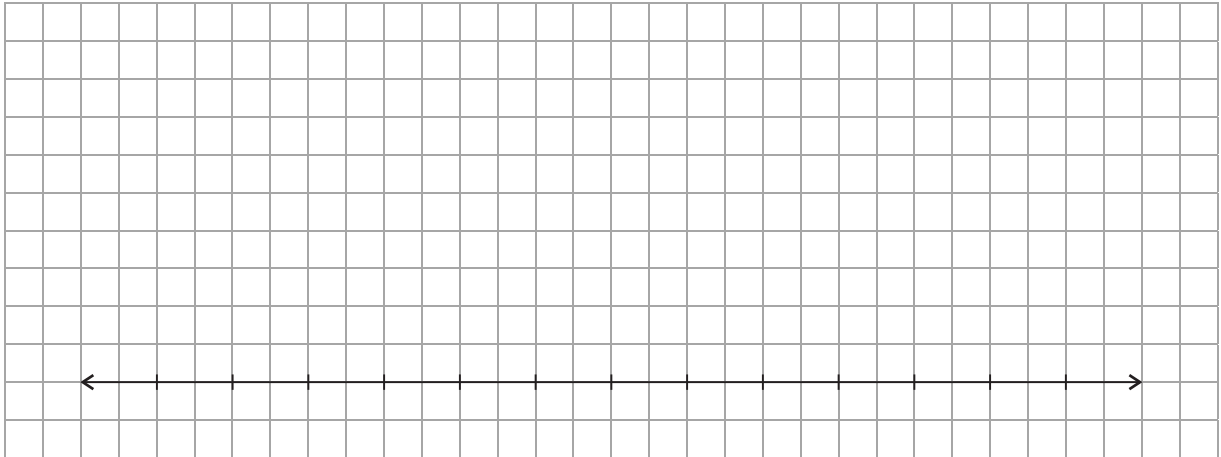
Note: exclude the length of time Emma spent at the meeting when calculating the average speed.

[illegible]

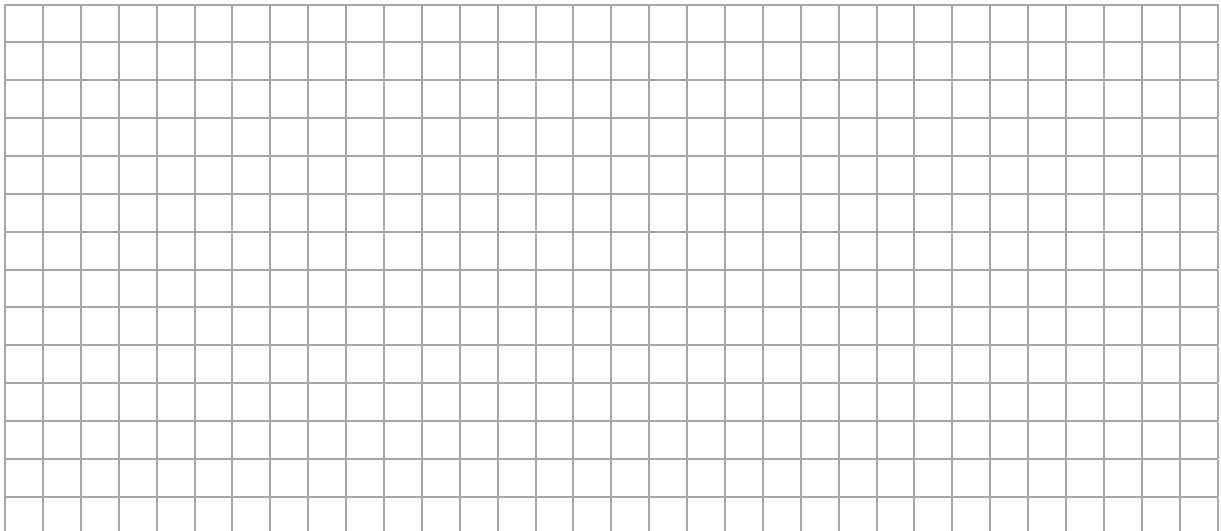
Question 11**(Suggested maximum time: 15 minutes)**

- (a) Solve the following inequality and graph your solution on the number line.

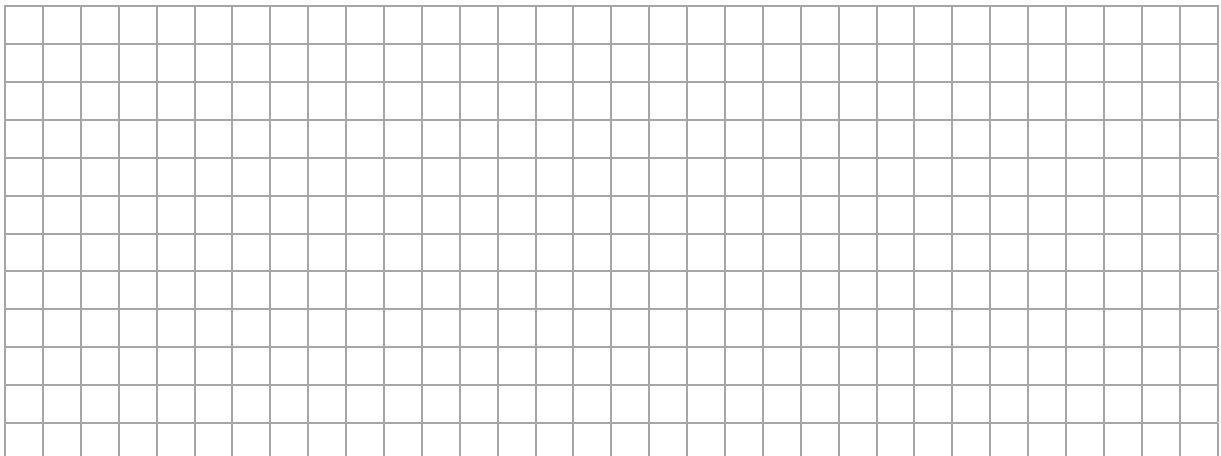
$$-2 < 7 - 2x \leq 9, x \in \mathbb{Z}.$$



- (b) Divide $2x^3 + 7x^2 + 2x - 8$ by $x + 2$.



- (c) Simplify $\frac{6}{4x+3} + \frac{4}{5x-1}$.

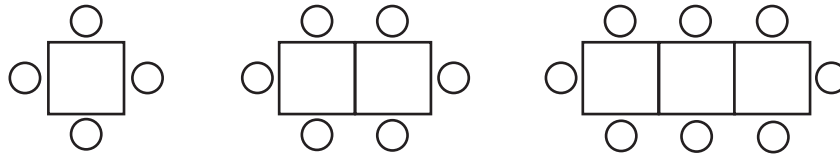


Question 12 (Suggested maximum time: 10 minutes)

Question 12 (Suggested maximum time: 10 minutes)

Customers at a restaurant can sit around square tables.

If a table is required for more than four people, two or more tables are joined together, as shown in the diagram below.



- (a)** What is the maximum number of customers that can sit around a row of five square tables?

[illegible]

- (b)** Using a table, a graph, or otherwise, write a formula to express C in terms of T , where C is the number of chairs and T is the number of tables.

[illegible]

A customer books a table for 26 people, requesting that they all sit together for dinner.

- (c)** How many tables are required so that they can all sit together?

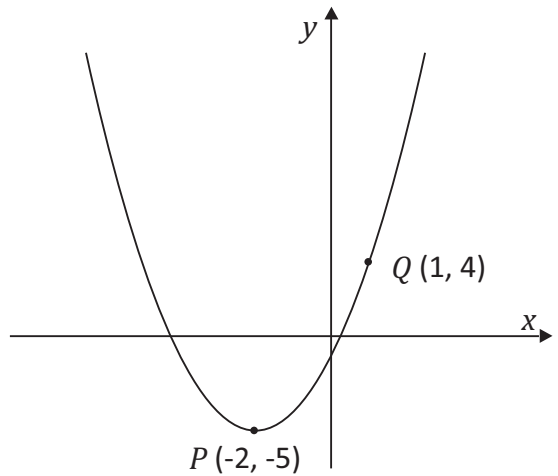
[illegible]

Question 13

(Suggested maximum time: 15 minutes)

The graph shown is given by the function $f(x) = x^2 + ax + b$, where a and $b \in \mathbb{Z}$, and $x \in \mathbb{R}$.

The points $P(-2, -5)$ and $Q(1, 4)$ are on the graph as shown.



- (a) Write two equations in a and b , using the coordinates of points P and Q .

- (b)** Solve the two equations, and hence find values for a and b .

[illegible]

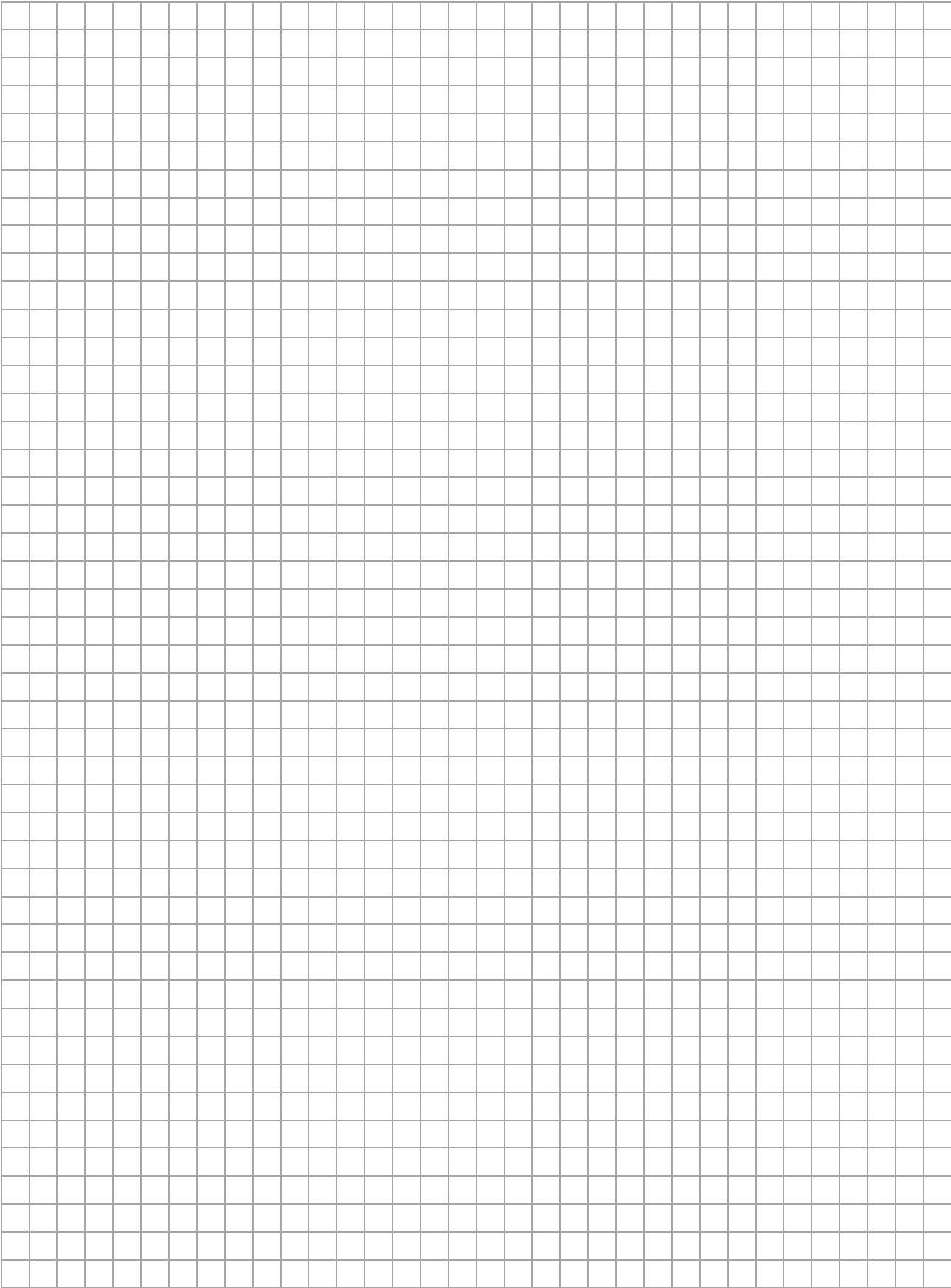
- (c)** What are the coordinates of the point where the graph cuts the y -axis?

Answer = (,)

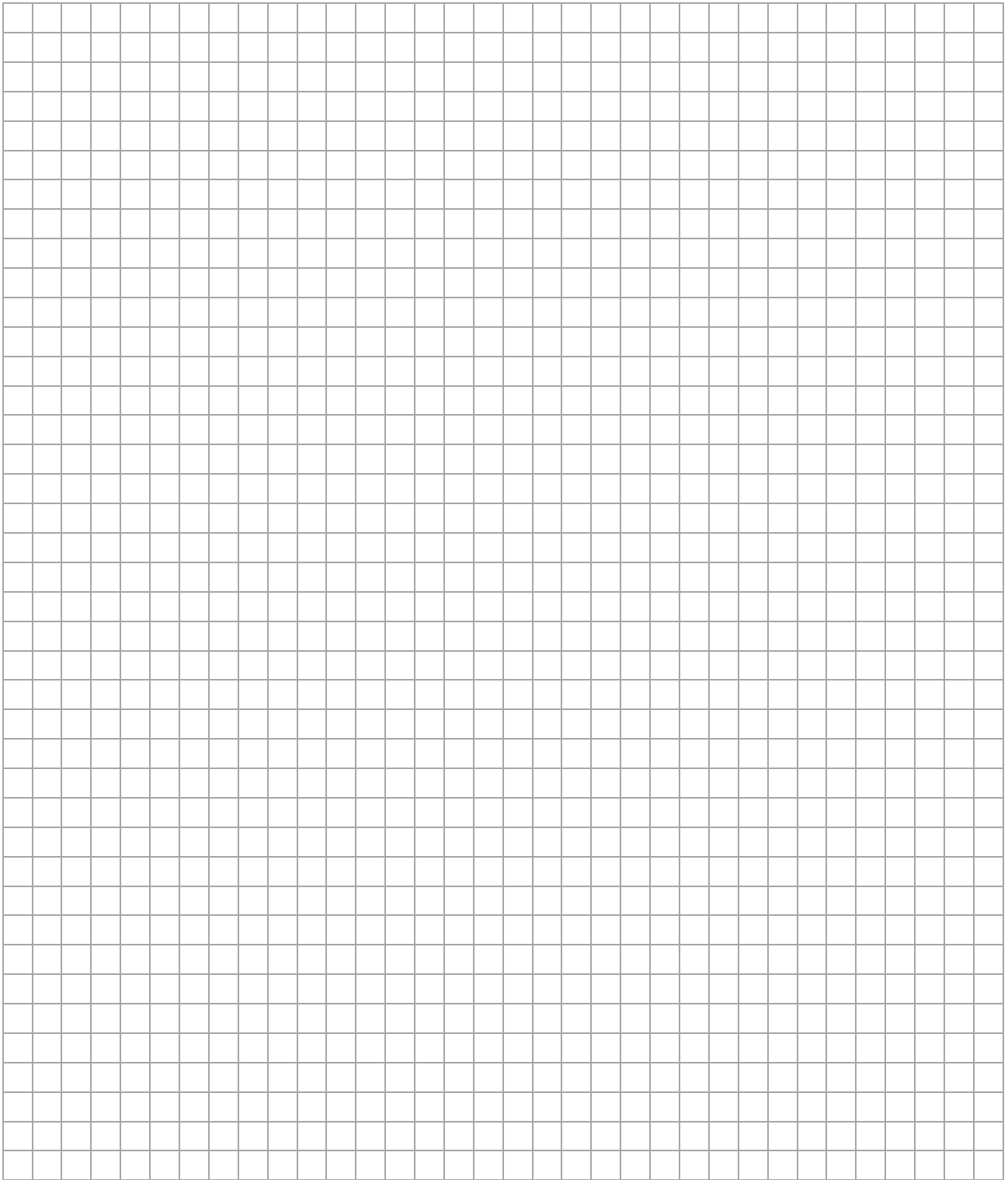
- (d)** Find the x -intercepts of the function, $y = x^2 + ax + b$ correct to one decimal place.
Hint: Use the quadratic equation formula.

[illegible]

You may use this page for extra work.



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Pre-Junior Certificate, 2018 – Higher Level

Mathematics – Paper 1

Time: 2 hours, 30 minutes

